

```
import numpy as np
import matplotlib.pyplot as plt
```

```
form_factor_params = [
    # Z,    a,      b1,      c,      q1,      b2,      q2,      b3,      q3,      b4,      q4
    {"Z": 1, "a": 3.566, "b1": -1.143, "c": -2.243, "q1": 0.20, "b2": 6.102, "q2": 0.6, "b3": 4.442, "q3": 0.90, "b4": 3.921, "q4": 15},
    {"Z": 2, "a": 3.264, "b1": 0.019, "c": -1.773, "q1": 0.30, "b2": 3.695, "q2": 0.7, "b3": 2.905, "q3": 1.25, "b4": 3.736, "q4": 8},
    {"Z": 3, "a": 3.010, "b1": 2.581, "c": -3.231, "q1": 0.25, "b2": 1.647, "q2": 0.5, "b3": 2.177, "q3": 1.50, "b4": 3.591, "q4": 8},
    {"Z": 4, "a": 6.343, "b1": 0.470, "c": -2.279, "q1": 0.20, "b2": 1.412, "q2": 0.5, "b3": 1.559, "q3": 2.50, "b4": 3.606, "q4": 8},
    {"Z": 5, "a": 6.625, "b1": 1.007, "c": -2.285, "q1": 0.25, "b2": 1.593, "q2": 0.5, "b3": 1.248, "q3": 2.50, "b4": 3.435, "q4": 8},
    {"Z": 6, "a": 7.366, "b1": 0.745, "c": -3.209, "q1": 0.25, "b2": 2.395, "q2": 0.5, "b3": 1.026, "q3": 2.50, "b4": 3.258, "q4": 8},
    {"Z": 7, "a": 8.657, "b1": 0.222, "c": -3.815, "q1": 0.25, "b2": 2.787, "q2": 0.5, "b3": 0.878, "q3": 2.50, "b4": 3.003, "q4": 8},
]
```

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form_factor_params1 = [
    {"Z": 8, "a": -2.038, "b1": 17.634, "c": 2.887, "q1": 0.50, "b2": 1.339, "q2": 0.839, "b3": 0.718, "q3": 7.0},
    {"Z": 9, "a": -30.997, "b1": 5.569, "c": 4.307, "q1": 0.25, "b2": 2.800, "q2": 0.656, "b3": 0.659, "q3": 7.0},
    {"Z": 10, "a": -8.137, "b1": 7.102, "c": 3.058, "q1": 0.13, "b2": 2.419, "q2": 0.768, "b3": 0.605, "q3": 7.0},
    {"Z": 11, "a": -1.108, "b1": 29.778, "c": 1.972, "q1": 0.40, "b2": 2.157, "q2": 0.902, "b3": 0.555, "q3": 7.0},
    {"Z": 12, "a": -0.599, "b1": 51.200, "c": 1.836, "q1": 0.20, "b2": 1.884, "q2": 1.024, "b3": 0.512, "q3": 7.0},
    {"Z": 13, "a": -0.618, "b1": 61.112, "c": 1.926, "q1": 0.25, "b2": 1.717, "q2": 1.132, "b3": 0.476, "q3": 7.0},
    {"Z": 14, "a": -0.929, "b1": 45.786, "c": 2.093, "q1": 0.25, "b2": 1.538, "q2": 1.285, "b3": 0.442, "q3": 7.0},
    {"Z": 15, "a": -0.998, "b1": 45.579, "c": 2.055, "q1": 0.30, "b2": 1.425, "q2": 1.403, "b3": 0.413, "q3": 7.0},
    {"Z": 16, "a": -1.457, "b1": 33.964, "c": 2.154, "q1": 0.30, "b2": 1.321, "q2": 1.581, "b3": 0.373, "q3": 10.0},
    {"Z": 17, "a": -2.082, "b1": 25.555, "c": 2.245, "q1": 0.30, "b2": 1.268, "q2": 1.674, "b3": 0.354, "q3": 10.0},
    {"Z": 18, "a": -2.779, "b1": 19.784, "c": 2.290, "q1": 0.40, "b2": 1.084, "q2": 1.889, "b3": 0.338, "q3": 10.0},
    {"Z": 19, "a": -1.500, "b1": 29.714, "c": 2.008, "q1": 0.40, "b2": 1.105, "q2": 1.922, "b3": 0.319, "q3": 10.0},
    {"Z": 20, "a": -0.966, "b1": 50.494, "c": 1.901, "q1": 0.50, "b2": 0.963, "q2": 2.178, "b3": 0.300, "q3": 10.0},
    {"Z": 21, "a": -1.083, "b1": 50.773, "c": 1.873, "q1": 0.50, "b2": 0.952, "q2": 2.248, "b3": 0.287, "q3": 10.0},
    {"Z": 22, "a": -1.300, "b1": 39.583, "c": 1.852, "q1": 0.60, "b2": 0.838, "q2": 2.463, "b3": 0.276, "q3": 10.0},
    {"Z": 23, "a": -1.477, "b1": 37.779, "c": 1.816, "q1": 0.60, "b2": 0.842, "q2": 2.501, "b3": 0.265, "q3": 10.0},
    {"Z": 24, "a": -2.451, "b1": 22.944, "c": 1.862, "q1": 0.60, "b2": 0.819, "q2": 2.636, "b3": 0.252, "q3": 100.0},
    {"Z": 25, "a": -1.962, "b1": 28.845, "c": 1.756, "q1": 0.70, "b2": 0.753, "q2": 2.792, "b3": 0.243, "q3": 10.0},
    {"Z": 26, "a": -1.577, "b1": 44.670, "c": 1.651, "q1": 0.80, "b2": 0.679, "q2": 3.147, "b3": 0.226, "q3": 10.0},
    {"Z": 27, "a": -2.522, "b1": 22.906, "c": 1.697, "q1": 0.80, "b2": 0.659, "q2": 3.244, "b3": 0.219, "q3": 10.0},
    {"Z": 28, "a": -3.013, "b1": 19.358, "c": 1.692, "q1": 0.80, "b2": 0.656, "q2": 3.309, "b3": 0.212, "q3": 10.0},
    {"Z": 29, "a": -4.405, "b1": 14.344, "c": 1.728, "q1": 0.80, "b2": 0.651, "q2": 3.370, "b3": 0.205, "q3": 10.0},
    {"Z": 30, "a": -3.694, "b1": 16.277, "c": 1.634, "q1": 0.90, "b2": 0.603, "q2": 3.533, "b3": 0.203, "q3": 10.0},
    {"Z": 31, "a": -3.245, "b1": 17.357, "c": 1.580, "q1": 0.90, "b2": 0.612, "q2": 3.541, "b3": 0.200, "q3": 10.0},
    {"Z": 32, "a": -2.415, "b1": 24.695, "c": 1.491, "q1": 1.00, "b2": 0.576, "q2": 3.754, "b3": 0.190, "q3": 10.0},
    {"Z": 33, "a": -2.189, "b1": 29.184, "c": 1.457, "q1": 1.00, "b2": 0.579, "q2": 3.791, "b3": 0.187, "q3": 10.0},
    {"Z": 34, "a": -1.357, "b1": 65.926, "c": 1.344, "q1": 1.25, "b2": 0.491, "q2": 4.339, "b3": 0.175, "q3": 15.0},
    {"Z": 35, "a": -1.391, "b1": 66.774, "c": 1.329, "q1": 1.25, "b2": 0.490, "q2": 4.436, "b3": 0.169, "q3": 15.0},
    {"Z": 36, "a": -1.432, "b1": 68.070, "c": 1.320, "q1": 1.25, "b2": 0.486, "q2": 4.543, "b3": 0.164, "q3": 15.0},
    {"Z": 37, "a": -1.387, "b1": 49.729, "c": 1.333, "q1": 1.25, "b2": 0.467, "q2": 4.706, "b3": 0.159, "q3": 15.0},
    {"Z": 38, "a": -1.012, "b1": 78.049, "c": 1.306, "q1": 1.25, "b2": 0.470, "q2": 4.779, "b3": 0.154, "q3": 15.0},
    {"Z": 39, "a": -0.947, "b1": 92.565, "c": 1.294, "q1": 1.25, "b2": 0.468, "q2": 4.831, "b3": 0.151, "q3": 15.0},
]
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{
  "Z": 40, "a": -0.960, "b1": 96.320, "c": 1.285, "q1": 1.25, "b2": 0.466, "q2": 4.895, "b3": 0.148, "q3": 15.0},
  {"Z": 41, "a": -0.184, "b1": 297.380, "c": 1.241, "q1": 1.25, "b2": 0.478, "q2": 4.876, "b3": 0.145, "q3": 15.0},
  {"Z": 42, "a": 1.157, "b1": -0.320, "c": 1.275, "q1": 2.00, "b2": 0.375, "q2": 5.653, "b3": 0.136, "q3": 15.0},
  {"Z": 43, "a": 1.157, "b1": -0.320, "c": 1.275, "q1": 2.00, "b2": 0.363, "q2": 5.815, "b3": 0.133, "q3": 15.0},
  {"Z": 44, "a": 1.157, "b1": -0.319, "c": 1.265, "q1": 2.00, "b2": 0.358, "q2": 6.006, "b3": 0.117, "q3": 15.0},
  {"Z": 45, "a": 1.141, "b1": -0.488, "c": 1.269, "q1": 2.00, "b2": 0.411, "q2": 5.624, "b3": 0.127, "q3": 15.0},
  {"Z": 46, "a": 1.141, "b1": -0.488, "c": 1.269, "q1": 2.00, "b2": 0.400, "q2": 5.734, "b3": 0.124, "q3": 15.0},
  {"Z": 47, "a": 1.141, "b1": -0.488, "c": 1.269, "q1": 2.00, "b2": 0.379, "q2": 6.119, "b3": 0.117, "q3": 15.0},
  {"Z": 48, "a": 1.141, "b1": -0.488, "c": 1.269, "q1": 2.00, "b2": 0.71, "q2": 6.219, "b3": 0.114, "q3": 15.0},
  {"Z": 49, "a": -2.015, "b1": 59.177, "c": 1.308, "q1": 0.80, "b2": 0.782, "q2": 2.226, "b3": 0.326, "q3": 7.0},
  {"Z": 50, "a": -1.957, "b1": 61.985, "c": 1.302, "q1": 0.80, "b2": 0.770, "q2": 2.265, "b3": 0.324, "q3": 7.0}
}
```

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]
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```
def F(z,q):
    if z>50:
        raise ValueError("nur bis z <= 50 implementiert")

    if z <=7:
        for para in form_factor_params:
            if para["Z"] == z:
                if q < 0:
                    raise ValueError("q < 0")

                q1 = para["q1"]
                b1 = para["b1"]
                b2 = para["b2"]
                a = para["a"]
                c = para["c"]
                q2 = para["q2"]
                b3 = para["b3"]
                q3 = para["q3"]
                b4 = para["b4"]
                q4 = para["q4"]

                f11 = a * np.exp(-b1 * q) + (z-a) * np.exp(-c*q)
                f12 = f11 * np.exp(-b2*(q1-q))
                f13 = f12 * np.exp(-b3 * (q2 - q))
                f14 = f13 * (q/q3)**b4

                if q <= q1:
                    return f11
                if q <= q2:
                    return f12
                if q <= q3:
                    return f13
                if q <= q4:
                    return f14
```

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        else:
            raise ValueError("q>q4")
else:
    for para in form_factor_params1:
        if para["Z"] == z:
            q1 = para["q1"]
            b1 = para["b1"]
            b2 = para["b2"]
            a = para["a"]
            c = para["c"]
            q2 = para["q2"]
            b3 = para["b3"]
            q3 = para["q3"]

            f21 = a * np.exp(-b1 * q) + (z-a) * np.exp(-c*q)
            f22 = f21 * np.exp(-b2*(q1-q))
            f23 = f22 * np.exp(-b3 * (q2 - q))

            if q <= q1:
                return f21
            if q <= q2:
                return f22
            if q <= q3:
                return f23
            else:
                raise ValueError("q>q3")

```